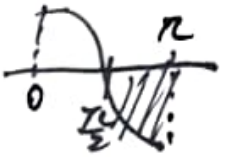
Ex. 8.3

Q 29

$$\int_{5\pi/6}^{\pi} \frac{\cos^4 x}{\sqrt{1-\sin x}}\,dx \cdot \frac{\sqrt{1+\sin x}}{\sqrt{1+\sin x}}$$



$$= \int_{5\pi/6}^{\pi} \frac{\cos^4 x\sqrt{1+\sin x}}{\sqrt{1-\sin^2 x}}\,dx \longrightarrow 1-\sin^2 x = \cos^2 x$$

$$\sqrt{1-\sin^2 x} = |\cos x| = -\cos x$$

$$= \int_{5\pi/6}^{\pi} -\cos^3 x\sqrt{1+\sin x}\,dx$$

$u = 1+\sin x$

$du = \cos x\,dx$

$x = \pi \Rightarrow u = 1+\sin\pi = 1$

$x = \frac{5\pi}{6} \Rightarrow u = \frac{3}{2}$

$$= \int_{3/2}^{1} -\cos^2 x\sqrt{u}\,du$$

$$= \int_{1}^{3/2} \cos^2 x\sqrt{u}\,du$$

$$= \int_{1}^{3/2} (-u^2+2u)\underbrace{\sqrt{u}}_{u^{\frac{1}{2}}}\,du$$

$$= \int_{1}^{3/2} -u^{5/2}+2u^{3/2}\,du$$

$$= \left[-\frac{2}{7}u^{7/2} + \frac{4}{5}u^{5/2}\right]_1^{3/2}$$

~~$\cos^2 x = 1-\sin^2 x$~~

$u = 1+\sin x$

$\cos^2 x = 1-\sin^2 x$

$= 1-(u-1)^2$

$= 1-u^2+2u-1$

$= -u^2+2u$

Ex. 8.4

Q48 $\displaystyle\int \frac{\sqrt{x-2}}{\sqrt{x-1}}\,dx$

$\sqrt{t^2-1} \nearrow \sec\theta = t$

$\displaystyle = \int \frac{\sqrt{u-1}}{\sqrt{u}}\,du \qquad u = x-1$

① Multiply by $\dfrac{\sqrt{u-1}}{\sqrt{u-1}}$

$$\text{Integral} = \int \frac{(u-1)}{\sqrt{u(u-1)}}\,du$$

$$= \int \frac{u-1}{\sqrt{u^2-u}}\,du$$

$$= \int \frac{u}{\sqrt{u^2-u}}\,du - \int \frac{1}{\sqrt{u^2-u}}\,du$$

Completing the square

② $t = \sqrt{\dfrac{u-1}{u}} \Rightarrow t^2 = \dfrac{u-1}{u}$

$= 1 - \dfrac{1}{u}$

$\Rightarrow 2t\,dt = \dfrac{1}{u^2}\,du$

$du = 2tu^2\,dt$

$$\int \sqrt{\frac{u-1}{u}}\, du = \int t \cdot 2t\, u^2\, dt$$

$$= \int 2t^2 \underline{u^2}\, dt$$

$$= \cancel{\frac{2}{u}}$$

$$t^2 u = u - 1$$

$$\cancel{\frac{u-1}{u}}$$

$$t^2 u^2 = \underline{u(u-1)}$$

not good

③ Let $t = \sqrt{u} \Rightarrow dt = \frac{1}{2\sqrt{u}}\, du$

$$\int \sqrt{\frac{u-1}{u}}\, du = \int \frac{\sqrt{u-1}}{\sqrt{u}}\, du$$

$$= \int 2\sqrt{t^2-1}\, dt$$

$t = \sec\theta$

Ex. 8.2

Q53 $\int (1+2x^2)e^{x^2}\,dx$

$u = x^2$

$du = 2x\,dx$

$2x^2e^{x^2}$

$\downarrow$

xe^u

$\uparrow$

$\int ue^u$

$$= \int 1\cdot e^{x^2}\,dx + \int 2x^2e^{x^2}\,dx$$

$$\int 1\cdot e^{x^2}\,dx = xe^{x^2} - \int x\cdot(2xe^{x^2})\,dx$$

$$= xe^{x^2} - \int 2x^2e^{x^2}\,dx$$

Can be cancelled

$$\therefore \int (1+2x^2)e^{x^2}\,dx = xe^{x^2} + C$$

Q54 $\int \frac{xe^x}{(x+1)^2}\,dx$

$u = x+1$

$du = dx$

$$= \int \frac{(u-1)e^{u-1}}{u^2}\,du$$

$$= \int u^{-1}e^{u-1}\,du - \int \frac{e^{u-1}}{u^2}\,du$$

$$= u^{-1}e^{u-1} - \int e^{u-1}\cdot\left(-\frac{1}{u^2}\right)du - \int \frac{e^{u-1}}{u^2}\,du$$

② $\displaystyle\int \frac{xe^x}{(1+x)^2}\,dx$

$$= \int \underbrace{xe^x}_{\text{der.}} \cdot \underbrace{\frac{1}{(1+x)^2}}_{\text{int.}}\,dx$$

$$= xe^x \cdot \left(-\frac{1}{1+x}\right) - \int (xe^x + e^x) \cdot \left(-\frac{1}{1+x}\right) dx$$

(by part)

Ex. 8.3

$\tan^2 x = \sec^2 x - 1$

Q42 $\displaystyle\int \boxed{\tan^4 x}\, \sec^3 x\, dx$

$$= \int (\sec^2 x - 1)^2 \sec^3 x\, dx$$

$$= \int (\sec^4 x - 2\sec^2 x + 1)\sec^3 x\, dx$$

$$= \int \sec^7 x - 2\sec^5 x + \sec^3 x\, dx$$

Solve $\int \sec^3 x\,dx$, $\int \sec^5 x\,dx$, $\int \sec^7 x\,dx$
Using by part.